Self Without a Transcript

Reading a preference and acting on a conditional rule

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Abstract

Can a model carry a rule from a self-description into a later decision, when the original words are gone? We separate reading a distinction, using it and updating it. On eight known topics, a probe of frozen Gemma states identified whether the first action was accepted or rejected on 2,043/2,048 rephrased records. Recovering the identity of the preferred action succeeded on 1,311/2,048. We then trained two literal readers from the same weights to apply a two-flag policy. One practised final actions, the other also practised flag reading, conjunction and branch selection. Under joint changes in names, grammar and questions, component training gave 90.5% correct direct actions versus 91.0% for final-action training. When the first call supplied only the conjunction bit and the second received that bit and the original policy text, the corresponding accuracies were 87.3% and 75.5%. The registered direct-transfer benefit and full intervention criterion both failed. This finite-domain contrast identifies the interface on which a descriptive gain appeared and specifies a test for a future learned two-step update.

1 A rule for choosing a route

A walker says, "I take the river path when signals Alder and Birch are both on. Otherwise I take the ridge path." Tomorrow Alder is on and Birch is off. The walker should take the ridge path. Now suppose the original sentence is gone. Could a model work from its current representation, expand the rule into cases, turn those cases back into a rule, and still choose the ridge path when asked tomorrow? What would change if it had to put its intermediate state into words?

Form of the same policy	What the model would need to preserve
Original description	Take the river path when both signals are on. In every other situation take the ridge path.
Exhaustive cases	Off/off, ridge. Off/on, ridge. On/off, ridge. On/on, river.
Reintegrated rule	Take the ridge path by default. Switch to the river path exactly when both signals are on.

The opposed policy exchanges river and ridge in every row. Exchanging the order of sentences changes neither policy. These transformations make the research question testable, but they also create an ambiguity. A model that simply copies the initial policy can give the correct final answer. Evidence for a useful update must therefore identify what the intermediate state supports, whether replacing it changes the later choice, and what is lost when an update is omitted.

We test prerequisites for that experiment. First we ask which relationship a question-free hidden state preserves when the actions are rephrased. Then we ask whether changing a source can change a subsequent model answer. Finally we compare two ways of teaching a model to apply

a conditional policy. In this last experiment an explicit one-bit interface lets us change a single intermediate decision and observe the resulting action. The first-person frame is a way to supply a policy. A claim specific to self-attribution would also require matched descriptions of another agent and impersonal rules.

2 What is read, and by whom?

Reading a state. An encoder reads a description before any evaluation question appears. We retain eight vectors at a specified layer and call this state $E(s)$. A source probe learns a score from these vectors to predict a feature of the description. A high probe score establishes that the fitted reader can recover that feature from one state. In a separate test, we inject a state into a task prompt and read the language model's own next token. We call this the *native answer*. An *opposed donor* supplies the state of the other policy while the question and its displayed options stay fixed.

Reading a choice. Every binary action question is presented twice, with the options exchanged. Native correctness requires the most likely token over the full vocabulary to be the requested letter. Other tokens count as errors. A strict unit in the conditional experiments fixes topic, flag-name pair, wording and representation stage. It passes only when both policies are correct in all four worlds, in every genuine source order and in both label orders. For example, a reader that always follows the last clause can score many individual decisions and still fail the entire unit. The original description and the table each have two orders. The reintegrated rule has one realization per wording family.

Reading across updates. The eventual comparison has three paths. *Rewrite in words* (`T`) passes generated text to the next step. *Continue in slots* (`L`) passes the updated hidden state. *Say it, encode it, then continue* (`B`) verbalizes that same intermediate state and encodes the words before the next update. The encoder, verbalizer and updater have internal identifiers E, D and G. The learned operation sequence is expansion into cases (G_1) followed by reintegration into a rule (G_2). The primary language intervention shares the same first-stage state across the last two paths. Generation proceeds to native end of sequence when context permits, with the actual stopping reason and any censored text retained.

Earlier Mistral-7B-Instruct-v0.3 and Gemma-2-2B-IT probes explain the ordering of our tests. Both models answered 24/24 preference questions from literal descriptions. Twenty-four distinct updated slot tensors nevertheless produced one verbalization per checkpoint, and all 12 re-encoded donor contrasts tied. Replaying states from positions that had already processed both options could reproduce native lexical choices. A recurrence needs a state formed *before* the question. The original tests of uncalibrated transport, output positions and stopping conditions are indexed in the supplement. Development experiments 1–8

Related work

The Quine Test studies whether descriptions reproduce preference profiles in fresh instances [1]. Propositional probes distinguish facts and their relational bindings in context [2]. Our verdict probe asks which of two familiar judgment roles survives a change in the action words. The donor and branch exchanges address the separate question of causal use. Coconut trains reasoning through continuous hidden-state feedback [5], and recurrent-depth models allocate repeated computation within a network [4]. Natural Language Autoencoders provide a learned activation-to-language-to-activation interface [3]. The planned language insertion compares later use on the two paths from the very same intermediate state.

3 A frozen state preserves the order of familiar verdicts

Consider a two-record description that accepts the river route and rejects the ridge route. Hold the two actions and their mention order fixed, then exchange the verdicts. The two descriptions contain the same tokens and the same slot positions, yet which action is accepted changes. A reader using only one question-free state can be tested on each member independently.

We froze Gemma-2-2B-IT and trained a linear score with a fitted offset on 4,608 such records. The readout predicted whether the first record accepted its action. A second readout predicted which action was mentioned first within an inferred topic. Combining the two predictions selected the preferred action. Evaluation used 64 unfamiliar action expressions across eight known topics, two familiar verdict vocabularies, both field placements, both action orders and both policies. The 2,048 states are repeated realizations of 16 policies within those topics.

Question asked of one held-out state	Correct	Comparison or consequence
Is the first action accepted?	2,043/2,048	Shuffling the training role supervision gave 1,022/2,048.
Are both members of an opposed pair read correctly?	1,019/1,024 pairs	Each member was read separately.
Which specific action is preferred?	1,311/2,048	Topic routing and action identity also have to be correct.

The role result passed its registered criterion in all four combinations of verdict vocabulary and field placement. A fit without extra weight on opposed pairs was equally accurate on the role. Each description contains one acceptance and one rejection, so a cue in either record can reveal their order. The result establishes a recoverable *order of familiar verdicts* under action rephrasing. The complete-policy errors locate the next difficulty. In 527 of the 737 errors, topic and verdict order were correct but the reader identified the wrong first action. Another 206 involved an incorrect topic and four identified the right action but assigned it the wrong verdict within the correct topic. Probe, shuffled control and error partition

4 When a source state changes a model answer

A separate intervention asks whether a source can drive a native choice. We first established that eight vectors derived from explicit option words could influence a frozen Gemma reader. We then fitted a map from a question-free source encoding into this working input space. At test time the map received one source state. The downstream question contained the displayed alternatives, and the model produced its ordinary next-token scores. Source and recipient were frozen during fitting of the map. Word-derived input control

On the 32 fitting states the mapped sources gave 64/64 native choices. Opposed donors reduced the score against the original gold to 0/64 and reversed the A/B margin on all 64 judgments. The fitting mean and zero inputs each gave 32/64. Wording of the question transferred better than wording of the source. The same states gave 62/64 under a withheld question wording, whereas changed source wording yielded 57/128 and two new topics yielded 30/64. This establishes causal use of the supplied preference on its fitting bank. General conditional reading requires a further operation. The same stored policy now has to yield *different* actions depending on the later world. Source exchange and transfer panels

We tested that operation on the eight known action topics by supplying each original description, case table and reintegrated rule independently. These were experimenter-authored teachers.

None was generated by a learned update. Native choices from ordinary slots were 127/256, 128/256 and 64/128, respectively, against balanced zero-state counts of 128/256, 128/256 and 64/128. Literal text gave 219/256, 215/256 and 88/128. Because even the original text left errors, the next experiment trained a reader on text before admitting slot learning. Independent teacher controls

Literal training and one fixed continuation raised fitting accuracy to 3,751/3,840 (97.7%). The corresponding strict counts were 23/48 for the original description, 28/48 for the table and 47/48 for the rule. Only the rule cleared the fitting requirement of 95% native and 90% strict. New names, source wording and questions gave 923/1,280 native choices. Of 89 fitting errors, 85 returned B when A was correct. This discrepancy motivates the closing experiment. It keeps both label orders, separates changes in names, grammar and questions, and asks where the decision fails when its parts can be tested independently. Literal curriculum · Margin continuation and label-order diagnostic

5 From two flags to an action

We compared two training curricula at equal update, target and padded-token budgets. Both began from the same saved literal reader on a frozen Gemma backbone. *Practise the final choice* (`direct`) spends every target on choosing the action from the policy and the two current flag values. *Learn the parts of the choice* (`parts`) splits its targets equally among reading one flag, deciding whether both are on, selecting an action from a supplied branch value, and answering the ordinary action question. Both arms train the same rank-16 query/value adapters in the final four layers. They use the same ordered source units, loss and fresh optimizer settings for 384 updates of 32 answers. Alongside full-vocabulary cross entropy, both include a gold-letter margin and a penalty for a common A/B bias under a swap of the displayed options.

We fixed the final weights before scoring any new panel. Familiar or new indicator names, source grammar and question wording form eight crossed cells. Each cell includes 1,280 native decisions on 160 descriptions. The primary comparison uses the cell in which all three factors changed. A training benefit requires a net gain of at least 64 decisions (five percentage points), gains in at least five of eight topics, and no loss in the total number of strict units. This paired test concerns the allocation of a fixed training budget. The historical absolute thresholds for using the literal reader are reported separately. Prospective design and decision rules

The secondary endpoint tests an explicit intermediate state. The first call receives only the two readings and answers "Are both on?" The answer's displayed label maps to a single Boolean value. The second call receives the policy and a fixed verbal rendering of that bit. It has no access to the individual readings. We also complement the bit at fixed policy or exchange the policy at fixed bit. Both interventions should reverse the final action. The two calls have independently swapped answer labels. Identical calls are reused, and an invalid first answer counts against every path that needs it. A separate true-bit oracle localizes errors after the world test. This experiment uses a hand-specified one-bit message and retains the original policy in the second call. For the walker, a false bit should lead to the ridge and exchanging it for true should send the walker to the river. A learned hidden-state updater would also have to carry the policy itself.

Both arms received 12,288 labels, 3,072 training forwards and 1,747,152 padded training tokens. The number of unmasked tokens was 1,747,152 for final-choice training and 1,369,704 for component training. On the jointly new panel the component arm recovered 19 answers that the final-choice arm missed and lost 25 that it got right. The net was $-6/1{,}280$ with gains in four of eight topics. The primary benefit criterion failed. Changing names alone left the final-choice arm at 1,276/1,280, its familiar count. Changing source grammar alone reduced it to 1,209/1,280. The component arm

showed a similar pattern, scoring 1,272 and 1,199 respectively. These contrasts concern the specified templates.

Jointly new stage	Final choice, native	Parts, native	Final choice, strict	Parts, strict
Description	460/512	459/512	1/16	1/16
Exhaustive cases	470/512	468/512	3/16	0/16
Reintegrated rule	235/256	232/256	6/16	3/16
All three	1,165/1,280	1,159/1,280	10/48	4/48

For the one-bit route, both arms read every individual flag correctly (32/32). The final-choice arm recognized the conjunction on 13/16 native questions, the component arm on 15/16. In the latter arm, when Nacre was on, Flint was off and A meant yes, the model answered A to the question of whether both were on. It correctly answered the same question with the labels reversed. That one error failed the per-order criterion. The second call can reuse the same bit across many policies, so 2,560 composed paths draw on just 16 conjunction judgments and 640 distinct second-call responses per arm.

Training	Predicted bit	True bit	Both bit settings correct	Strict units
Final choice	1,932/2,560	2,328/2,560	1,718/2,560	0/48
Learn parts	2,234/2,560	2,370/2,560	2,049/2,560	11/48

Component supervision improved the observed two-call score by 302 paths (11.8 percentage points) relative to the matched training control. Supplying the true bit increased this arm's count by a further 136, leaving action errors even with the world test solved. The 11 strict units comprised 3/16 descriptions, 2/16 tables and 6/16 rules. Correct original and opposed-policy answers occurred on 2,072/2,560 paths. The predeclared constructive criterion required at least 90% final correctness, 75% strict units and 90% correct paired reversals for both interventions at *each* stage, as well as 90% on each component under each label mapping. It failed. The observed two-call gain is a descriptive comparison under one schedule and one checkpoint. Its 87.3% accuracy also falls below the same arm's ordinary 90.5% accuracy. All eight panels, stage counts and native intervention records

6 From a checkable intermediate to learned recurrence

Training a reader to recognize the role of a phrase can make that role readable across new action words. A native reader can also use an intermediate decision where the interface explicitly supplies it. In the closing comparison, component practice improved performance on that particular two-call path while ordinary action transfer showed a net loss of six decisions. A larger recursive model would have to learn which distinction to retain, transmit it faithfully and exploit it in a new operation. The difference between the two paths motivates testing all three abilities separately as compute grows.

For an actual state update, first train a native reader on independently encoded descriptions, case tables and rules. Freeze its weights and scoring rule for independent confirmation on newly sealed policy groups. Then train the updater to generate the case state from the current state alone and to reintegrate it on the next call. On held-out descriptions test both intermediate and final reconstructions against the truth table. Exchange the intermediate with a same-meaning

paraphrase and with the opposed policy. Omit each operation in turn. Identity can preserve the final action, so a stage earns credit for doing useful work only when its specified distinction and later use survive these controls. Compare direct continuation with one language insertion from the identical saved intermediate. Record the extra decoding and re-encoding cost. Two-stage protocol

Scaling becomes an empirical question once that small route works. Hold out whole policy and operation families before training, including disjunctions, exceptions and rules with new actions. Increase training diversity, depth and model size in separate comparisons across multiple seeds within one model family. At each size compare learned recurrence to an ordinary final-answer model given the same training and inference compute, including all readouts and language conversions. Depths zero, one, two and four distinguish learning a single useful update from reusing it. To test a claim about broader intelligence, predeclare independent planning and language-reasoning tasks and retention controls, then estimate paired transfer effects by task family and seed. The current finite truth tables supply a way to decide whether to invest in those comparisons.

Supporting material

The development record and experiment key names each earlier experiment and preserves its run identifiers, results tables and links to the raw records. The reproduction instructions give the execution seals. The reference audit records source and version information.

References

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